

Project Design Phase

Problem – Solution Fit Template

Date	19 September 2026
Team ID	
Project Name	ShopEZ — Apparel E-Commerce Application
Maximum Marks	2 Marks

1. CUSTOMER SEGMENT(S)

Everyday clothing shoppers who want a fast, accurate online store where stock is trustworthy down to the size and color they need, and store administrators who need to manage the catalog, inventory, and orders without juggling separate tools.

2. JOBS-TO-BE-DONE / PROBLEMS

Shoppers need to register and log in securely, browse and filter a catalog by category, confirm real stock for a specific size/color combination, complete checkout with confidence, and track an order afterward. Admins need full product CRUD control (including variant-level stock), the ability to update order fulfillment status, and control over site-wide content like categories and the homepage banner.

3. TRIGGERS

A shopper landing on a product page and not being able to tell whether their size is actually in stock before adding it to cart; an admin discovering stock discrepancies after manually tracking inventory across spreadsheets or disconnected tools.

4. EMOTIONS: BEFORE / AFTER

Before: uncertain whether an order will arrive in the right size or color, and hesitant to complete checkout on a site that doesn't clearly confirm the order went through. After: confident in the purchase, able to track its status from “pending” through “delivered” without contacting support.

5. AVAILABLE SOLUTIONS

General-purpose storefront templates that track a single stock count per product rather than per variant; manual, spreadsheet-based inventory tracking maintained separately from the storefront; or fully custom builds that require significant engineering time to get right.

6. CUSTOMER CONSTRAINTS

Running the application requires Node.js (v18 or above), npm, and a MongoDB connection (local or a free MongoDB Atlas cluster). End users shopping on the deployed storefront need nothing beyond a modern web browser.

7. BEHAVIOUR

Shoppers browse and filter the catalog, then use the React client's Axios-based API layer to fetch product data, manage their cart, and place orders — with every action authenticated via a JWT attached automatically to each request.

8. CHANNELS OF BEHAVIOUR

8.1 ONLINE: The ShopEZ web storefront itself, reached directly or through search and social referral links.

8.2 OFFLINE: Word-of-mouth recommendations between shoppers; potential future in-store signage or QR codes directing customers to the online catalog.

9. PROBLEM ROOT CAUSE

Most small-to-mid storefront implementations track inventory at the whole-product level rather than per variant, and keep customer-facing and admin tooling as separate, disconnected systems — which is what causes stock inaccuracies and manual reconciliation overhead in the first place.

10. YOUR SOLUTION

ShopEZ addresses this with a single MERN application where variant-level stock, order-time snapshotting, and a unified admin panel (products, orders, and site settings) are all part of the same data model and codebase — eliminating the sync gap between what's shown as available and what's actually in stock.